

VizTR Platform

Complete Feature List - All Discussed Capabilities

1. Public Website

1.1 Marketing and Landing Pages

- 1.1.1 Hero section with brand messaging and visual identity
- 1.1.2 Services and capabilities showcase pages
- 1.1.3 Pricing page with plan tiers
- 1.1.4 Contact and inquiry forms

1.2 Public Project Portfolio / Showcase

- 1.2.1 Gallery grid of completed archviz projects
- 1.2.2 Individual project detail pages
- 1.2.3 Filter by project type and sector

1.3 AI Brief Submission Flow

- 1.3.1 Multi-step form for client project briefs
- 1.3.2 AI-powered concept render generation from brief inputs (Gemini API)
- 1.3.3 Email confirmation and brief summary delivery via Resend

1.4 XR Hub Page (/xr)

- 1.4.1 Dedicated XR landing and navigation hub
- 1.4.2 Links to all XR sub-routes: VR, AR, WebXR, Virtual Tours
- 1.4.3 301 redirects from legacy routes to canonical /xr/* paths
- 1.4.4 Navbar restructure to surface XR offerings

2. Client Portal

2.1 Authentication and Access Control

- 2.1.1 NextAuth-powered login with email and Google OAuth
- 2.1.2 Role-based access: client vs admin
- 2.1.3 Protected routes with session management

2.2 Client Project Dashboard

- 2.2.1 Per-client overview of all active projects
- 2.2.2 Project status tracking and milestone view
- 2.2.3 File and asset downloads per project

2.3 360 Degree Virtual Tour Viewer

- 2.3.1 Multi-resolution tile engine for smooth panning
- 2.3.2 Equirectangular panorama rendering
- 2.3.3 Hotspot navigation between spaces
- 2.3.4 Multi-user SSE live sessions within tours
- 2.3.5 Built on: Marzipano, Panolens.js (Three.js-native)

Note: Supports both desktop and mobile browsers

2.4 WebAR PBR Configurator

- 2.4.1 Physically Based Rendering material support
- 2.4.2 Real-time material/finish switching (floors, walls, surfaces)
- 2.4.3 AR placement via device camera (markerless)

- 2.4.4 HDRI studio lighting for material preview
- 2.4.5 Measurement overlays in AR view
- 2.4.6 Share or snapshot configured view
- 2.4.7 Built on: Google model-viewer, AR.js, pmndrs/drei (PBR material configurator)

2.5 VR Experience Viewer

- 2.5.1 Immersive VR mode for compatible headsets
- 2.5.2 Fallback 3DOF viewer on unsupported devices
- 2.5.3 Built on: A-Frame

2.6 WebXR Viewer

- 2.6.1 Mixed Reality (MR) mode for passthrough devices
- 2.6.2 Full WebXR API integration
- 2.6.3 Built on: Babylon.js

2.7 Multi-User Live Sessions

- 2.7.1 Real-time co-presence via SSE (Server-Sent Events)
- 2.7.2 Shared viewpoint synchronization across clients
- 2.7.3 Live annotation and pointer tools

3. Admin Panel

3.1 Admin Dashboard

- 3.1.1 KPI overview: active projects, sessions, revenue
- 3.1.2 Recent activity feed

3.2 Project Management

- 3.2.1 Create, edit, and archive projects
- 3.2.2 Assign projects to client accounts
- 3.2.3 Upload and manage 3D assets, panoramas, AR models
- 3.2.4 Asset pipeline: gltf-transform CLI for upload-time compression
- 3.2.5 Draco geometry compression and KTX2 texture compression
- 3.2.6 Asset storage: Cloudflare R2 via CDN

3.3 Session Booking System

- 3.3.1 Calendar UI for scheduling live sessions
- 3.3.2 Prisma-backed booking records in PostgreSQL
- 3.3.3 Automated email confirmations via Resend
- 3.3.4 Session status: pending / confirmed / completed

3.4 Billing and Payments (Stripe)

- 3.4.1 Subscription plan management
- 3.4.2 One-time invoice and payment links
- 3.4.3 Webhook handling for payment events
- 3.4.4 Client billing history view

3.5 Visual Hotspot Editor

- 3.5.1 Drag-and-drop hotspot placement on panorama canvas
- 3.5.2 Hotspot types: navigation, info popup, media embed
- 3.5.3 Save and publish hotspot config per tour

3.6 CMS and AI Content Management

- 3.6.1 Content management for portfolio and project pages
- 3.6.2 AI brief processing dashboard (Gemini API integration)
- 3.6.3 Ten landing page theme presets (switchable)

4. UE5 Pixel Streaming Integration

4.1 Self-Hosted GPU Workstation Setup

- 4.1.1 Dedicated on-prem workstation with GPU for UE5 rendering
- 4.1.2 Unreal Engine 5 project configured for Pixel Streaming
- 4.1.3 PM2 process management for stream daemon

4.2 Signalling and Networking

- 4.2.1 Cirrus signalling server for WebRTC negotiation
- 4.2.2 Cloudflare Tunnel exposed at wss://stream.viztr.io
- 4.2.3 coturn TURN server for NAT traversal and relay fallback

4.3 Frontend Integration

- 4.3.1 Embedded in Next.js via @epicgames/epicpixelstreaming
- 4.3.2 Mounted on the /xr route
- 4.3.3 Responsive player component with fullscreen support
- 4.3.4 Input forwarding: mouse, keyboard, touch events to UE5

4.4 Stream Admin Panel

- 4.4.1 Start, stop, restart stream from admin UI
- 4.4.2 Active session viewer (who is connected)
- 4.4.3 Stream health and latency monitoring

5. Infrastructure and Tech Stack

5.1 Frontend Stack

- 5.1.1 Next.js 15 App Router with RSC (React Server Components) - 50+ routes
- 5.1.2 TypeScript throughout the codebase
- 5.1.3 Tailwind CSS v4 with shadcn/ui component library
- 5.1.4 Framer Motion for animations and transitions
- 5.1.5 Three.js for 3D rendering (tours, WebXR)
- 5.1.6 Desktop app: Next.js + Tauri
- 5.1.7 Mobile app: Next.js + Capacitor

5.2 Backend and API

- 5.2.1 Next.js API Routes - 100+ endpoints
- 5.2.2 Prisma ORM connected to PostgreSQL (migrated from SQLite)
- 5.2.3 NextAuth for session and authentication
- 5.2.4 Resend for transactional emails (permanent free tier)
- 5.2.5 Stripe SDK for payments and webhooks

5.3 Open-Source Library Foundations

- 5.3.1 Virtual Tours: Marzipano
- 5.3.2 WebAR: Google model-viewer
- 5.3.3 VR: A-Frame
- 5.3.4 MR (Mixed Reality): AR.js
- 5.3.5 WebXR: Babylon.js
- 5.3.6 Three.js-native Tours: Panolens.js
- 5.3.7 PBR Material Configurator: pmndrs/drei

5.4 Asset Pipeline

- 5.4.1 Draco geometry compression
- 5.4.2 KTX2 texture compression
- 5.4.3 gltf-transform CLI for upload-time processing

5.4.4 Cloudflare CDN and R2 for global asset delivery

5.5 AI Agent Stack (Internal - VizTR)

5.5.1 Layer 1: LangGraph - orchestration and state management

5.5.2 Layer 2: CrewAI - 12-agent specialist team

5.5.3 Layer 3: Hermes - inter-agent communication

5.5.4 Layer 4: AgentGPT - autonomous task execution

5.5.5 Layer 5: Universal MCP Connector

5.5.6 Layer 6: Ollama / OpenAI inference backend

5.6 Deployment and DevOps

5.6.1 Cloudflare Tunnel for secure stream exposure

5.6.2 PostgreSQL hosted database (production)

5.6.3 Mobile-first development via Termux on phone

5.6.4 Local GPU workstation for heavy AI inference via Cloudflare Tunnel

6. @RahulShips Content Co-Pilot Dashboard

6.1 Core Platform

6.1.1 Built on Next.js 15 with Gemini API

6.1.2 IndexedDB for local-first data persistence

6.2 Modules

6.2.1 Daily Brief - content priorities and schedule for the day

6.2.2 Hook Bank - store and retrieve tested hook formulas

6.2.3 Competitors Hub - track and analyse competitor channels

6.2.4 Weekly Calendar Strip - visual content calendar

6.2.5 Link Vault - saved references and research links

6.2.6 AI Studio - AI-assisted script and caption generation

6.3 Content Pillars Tracked

6.3.1 Architecture Viz + AI (Gold)

6.3.2 Trading + Systems (Cyber Green)

6.3.3 Vibe Coding (Cosmic Cyan)

6.3.4 Builder Journey (Purple)